EXP 29

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clc;
clear;
close all;

% GSM Communication System with TDMA
% Simulation and Analysis Using MATLAB

% Step 1: Define GSM parameters
numUsers = 8;            % 8 TDMA users
bitsPerUser = 10;        % Bits transmitted per user
SNR_dB = 15;             % Signal-to-noise ratio

% Step 2: Generate random binary data for each user
data = randi([0 1], numUsers, bitsPerUser);

disp('Original Data of 8 Users:');
disp(data);

% Step 3: Create TDMA frame
% Each user gets one time slot per frame
tdmaFrame = zeros(1, numUsers * bitsPerUser);

for k = 1:numUsers
   idx = (k-1)*bitsPerUser + (1:bitsPerUser);
   tdmaFrame(idx) = data(k,:);
end

% Step 4: BPSK modulation
txSignal = 2*tdmaFrame - 1;
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% Step 5: Add AWGN channel noise
signalPower = mean(txSignal.^2);
noisePower = signalPower / (10^(SNR_dB/10));
noise = sqrt(noisePower)*randn(size(txSignal));

rxSignal = txSignal + noise;

% Step 6: BPSK demodulation
rxBits = double(rxSignal > 0);

% Step 7: Recover data for each user
recoveredData = zeros(numUsers, bitsPerUser);

for k = 1:numUsers
    idx = (k-1)*bitsPerUser + (1:bitsPerUser);
    recoveredData(k,:) = rxBits(idx);
end

% Step 8: Calculate bit error rate
numErrors = sum(tdmaFrame ~= rxBits);
totalBits = length(tdmaFrame);
BER = numErrors / totalBits;

% Step 9: Display results
disp('Recovered Data of 8 Users:');
disp(recoveredData);

fprintf('SNR = %d dB\n', SNR_dB);
fprintf('Total Bits Transmitted = %d\n', totalBits);
fprintf('Number of Bit Errors = %d\n', numErrors);
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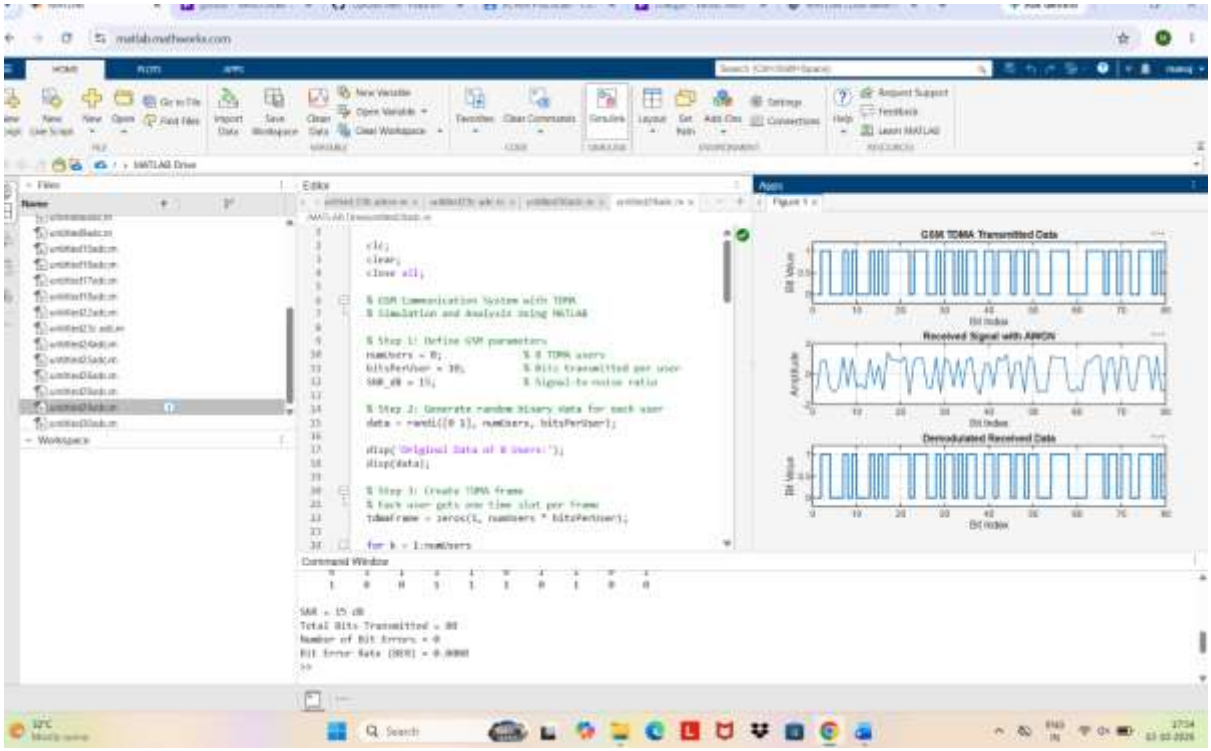
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fprintf('Bit Error Rate (BER) = %.4f\n', BER);

% Step 10: Plot results
figure;

subplot(3,1,1);
stairs(tdmaFrame, 'LineWidth', 1.5);
title('GSM TDMA Transmitted Data');
xlabel('Bit Index');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;

subplot(3,1,2);
plot(rxSignal, 'LineWidth', 1);
title('Received Signal with AWGN');
xlabel('Bit Index');
ylabel('Amplitude');
grid on;

subplot(3,1,3);
stairs(rxBits, 'LineWidth', 1.5);
title('Demodulated Received Data');
xlabel('Bit Index');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;
```


GSM TDMA Transmitted Data
Bit Value
Bit Index
Received Signal with AWGN
Amplitude
Bit Index
Demodulated Received Data
Bit Value
Bit Index